

Abstract

This project presents the design and simulation of a simplified GPU architecture implemented using Verilog. The proposed system focuses on the core principles of GPU execution by integrating essential components, including an instruction fetcher, decoder, scheduler, arithmetic logic unit (ALU), and a compute core. Instructions are fetched from memory, decoded into control signals, and executed through a lightweight execution pipeline that supports basic arithmetic and comparison operations. The compute core demonstrates multi-stage instruction flow and module interaction without advanced features such as pipelining or parallel warp execution. Through simulation-based verification, this project provides a clear and educational representation of how fundamental GPU components cooperate to perform computation, serving as a foundation for understanding more complex GPU architectures.